34º SIICUSP

A STRUCTURAL GENERALIZATION OF SUPERVISED LEARNING VIA CATEGORY THEORY

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OBJECTIVES

Supervised learning algorithms can be studied, from the perspective of Category Theory (Mac Lane, 1998), as transformations between data spaces that allow composition. This undergraduate research project aims to investigate the application of the approach presented by Fong *et al*. (2019) in the article*Backprop as functor*, which formulates backpropagation of Rumelhart et al. (1986) as a functor, for the structural representation of supervised learning. The investigation seeks to understand to what extent this structure can be extended, having as stages the implementation in categories, its generalization to multicategories and, as future work, its extension to PROPs.

METHODS AND PROCEDURES

The research is developed through the study of the construction presented by Fong *et al*. (2019) and its implementation in Haskell. In the categorical representation, learning algorithms are modeled as morphisms between data spaces, named learners, allowing their composition for the construction of more complex models. The implementation uses resources from the Haskell type system, including GADTs (Peyton Jones et al., 2006) and type families (Chakravarty et al., 2005), to represent parameter spaces as heterogeneous lists at the type level, as done by Kisselyov et al. (2004), allowing different parameter types to be associated with the same model.

From this structure, the research investigates the extension of the representation to multiple inputs. In the multicategorical generalization (Leinster, 2004), a morphism comes to represent a transformation of the form $(A_1, \ldots, A_n) \rightarrow B$.

The composition is then generalized to the substitution of operations on different inputs, according to the structure of a multicategory. The use of the same heterogeneous list representation for parameters and inputs introduces problems related to the association of types. Since ((p, q), r) and (p, (q, r)) receive distinct representations in the type system, unsafeCoerce is used to reconcile them.

RESULTS

The investigation resulted in the implementation of a categorical structure to represent supervised learning algorithms and its generalization to a multicategorical structure. The extension allows for the explicit representation of multiple inputs, incorporating arity into the morphism signature, while preserving the single-input case as a particular case.

As a functional validation of the implementation, a logistic regression model (Goodfellow et al., 2016) with a Binary Cross-Entropy (BCE) cost function was developed and applied to the Iris setosa and Iris versicolor classes of the Iris set (Fisher, 1936), characterizing a binary classification task. The model started with null parameters and, after training, converged to [1.0877; −1.5202; 1.8492; 1.8661; 0.2012].

The implementation achieved 100% accuracy in the subset used, a result consistent with the linear separability of the classes considered.

CONCLUSIONS

The research is ongoing and, to date, indicates that the categorical approach of Fong et al. provides an adequate framework for investigating supervised learning algorithms compositionally. Generalization to multicategories allows extending this representation to models with multiple inputs, maintaining composition as a structural element of learning.

As a continuation of the investigation, the intention is to study and implement PROPs (Mac Lane, 1965), allowing the representation of operations with multiple inputs and multiple outputs. This progression will allow assessing what level of categorical structure is necessary to represent different forms of composition in machine learning systems.

The authors declare no conflict of interest.

Contributions: Kauã Santana da Silva carried out the conception and implementation, the experiments, the analysis of the results, and the writing of the work. Alexandre Garcia de Oliveira provided scientific guidance, methodological discussion, and manuscript review.

Academic grant: This work did not receive an academic grant.

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